

PROJECT DOCUMENTATION

DIGITAL OHMMETER WITH 200 Ω
MAXIMUM RANGE

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Design of Industrial Devices and Measurement Systems 1

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1 Motivation

The assignment for the course "Design of Industrial Devices and Measurement Systems 1" is to develop a project that demonstrates knowledge and skills in designing the schematic and printed circuit board (PCB) for an electrical device. The project focuses on the development of an instrument for measuring resistance up to $200\ \Omega$. The goal of the project is to acquire and expand knowledge in the field of electronics and low-resistance measurement.

2 Introduction

The goal of this project is the development of a measuring instrument capable of measuring resistance in the range of 2 to $200\ \Omega$ with practical application in an industrial environment or laboratory. This documentation comprehensively covers the instrument's design, operational principles, and the selection of components integrated into the device itself. The approaches to challenges encountered during development will also be explained sequentially according to the stages of development.

IC	Consumption
MAX1491	980 μA
VIM 503-DP-RC-S-HV	10 μA
LM78M05_TO252	10 mA
3x OPA197IDGKT	3x 1 mA
INA826	200 μA
MAX9031AU	55 μA
Power LED	15 mA
Ind. k.s. LED	15 mA
AI-1223-TWT-5V-5-R	30 mA
Total:	72.245 mA

Table 1: Consumption estimation

Note: The specified values apply to $T = 25^\circ\text{C}$.

3 Schematic

3.1 Power supply

The device is designed to be powered by a single 9 V battery (**BT1**) that supplies the device with +9 V. The battery is located on a separate board with dedicated access on the enclosure for easier replacement. The battery board connects to the main board of the device via a two-pin power connector (**J3** and **J4**).

On the main board, after the connector, there is a power switch. The switch (**SW2**) is an SPST (single pole single throw) type, which meets the device's requirements given that it uses a unipolar power supply. The switch allows the user to turn the device on or off without the need to physically disconnect the battery.

The device features reverse polarity protection using a PMOS transistor (**Q1**) to protect the device in the event of an incorrectly connected battery, connector, etc. Under reverse battery polarity, the PMOS will not conduct, thereby preventing potential damage to the other components of the device. We use the PMOS FQP27P06 because it has a low $R_{ds(on)}$, thereby creating a negligible voltage drop—much lower than if we had used a diode. The V_{gs} is also critical for us, which for this transistor is ± 25 V, enabling reliable protection against a reverse-polarized 9 V battery.

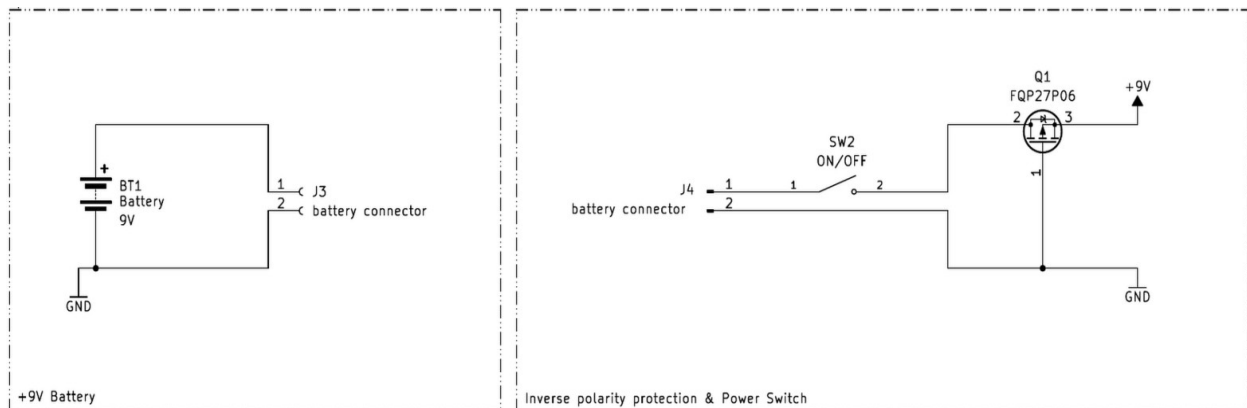


Figure 1: Battery & Inverse polarity protection

Further, the +9 V voltage is fed to the +5 V regulator (**U3**). This +5 V voltage is used to power all circuits in the device. For the +5 V regulator, we use the LM78M05. This regulator is key to providing a stable voltage for all device components. The regulator was chosen according to the power supply needs (+5 V) and the total current the device can draw. The LM78M05 can provide up to 500 mA of current, which satisfies the needs of this device and leaves headroom for safety. Diodes (**D1** and **D2**) serve to protect the regulator itself from reverse current flow.

Capacitors (**C9**, **C10**, **C11**, **C12**) serve to further stabilize the voltage and reduce the impact of environmental noise. Two electrolytic capacitors (**C9**, **C12**) and two ceramic block capacitors (**C10**, **C11**) are used.

At the output of the regulator, there is an indicator LED (**D3**) that serves to indicate the state of the device. Test points (**TP1** and **TP2**) allow easy access to the supply voltage for easier diagnostics.

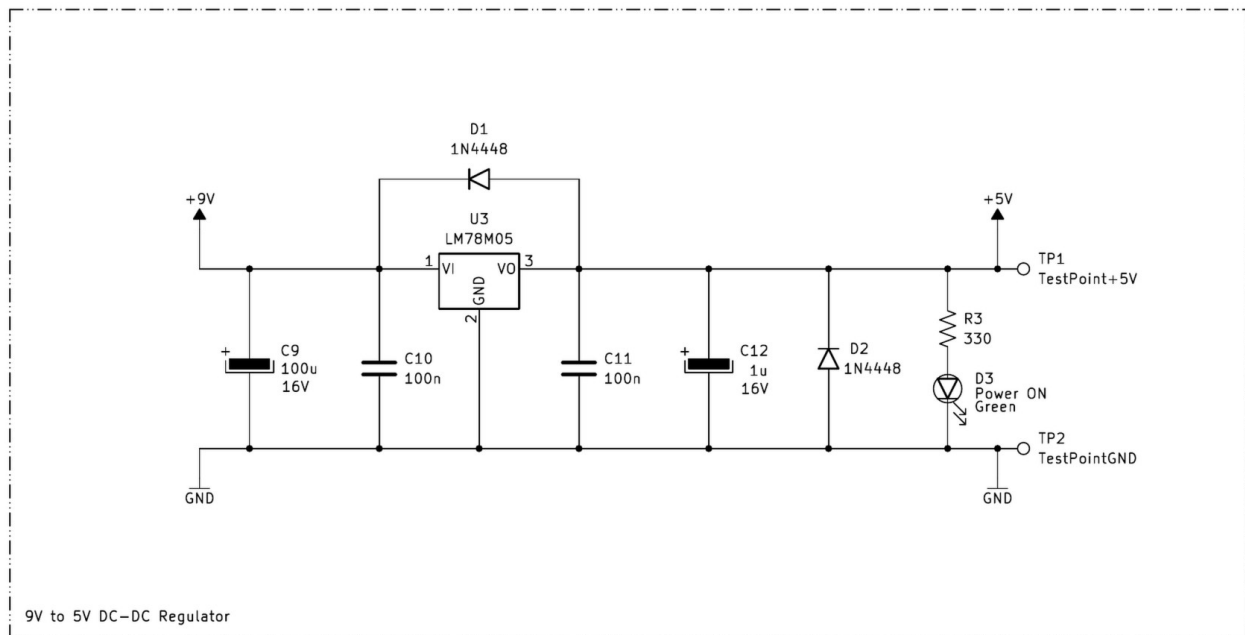


Figure 2: 9 V to 5 V DC-DC regulator

3.2 Current source

To design the ohmmeter, a constant current source is required. In order to determine the value of the resistance being measured, we must pass a known current through that resistance and then measure the voltage drop across it. The constant current source consists of two operational amplifiers (**U4** and **U5**), a PMOS transistor (**Q2**), and sense resistors (**R7**, **R9**, **R10**). This current source is designed to precisely provide the current value required for a given measurement. It is adjustable across ranges of 1 mA, 10 mA, and 100 mA, which correspond to the measurement ranges of this device (200 Ω , 20 Ω , and 2 Ω , respectively). Range selection is implemented using a DP3T switch (**SW3**), which selects the current range while simultaneously sending a signal to the main chip regarding the selected range—i.e., the position of the decimal point.

The current source features overvoltage protection and electrostatic discharge (ESD) protection, as well as a PTC fuse that protects against overcurrent. Overvoltage protection is realized using a pair of BAT54S diodes (**D4**) that clamp the voltage within the operating voltage range. ESD protection is implemented via an SMBJ5.0A TVS diode (**D5**), which protects the remaining components in the event of a surge. In the case of a fault current, a resettable PTC fuse (**F1**) with a hold current of up to 200 mA trips to isolate the circuit. The output of this current source is provided via S16N-PC jacks (**J5** and **J6**), which serve as the device's current terminals.

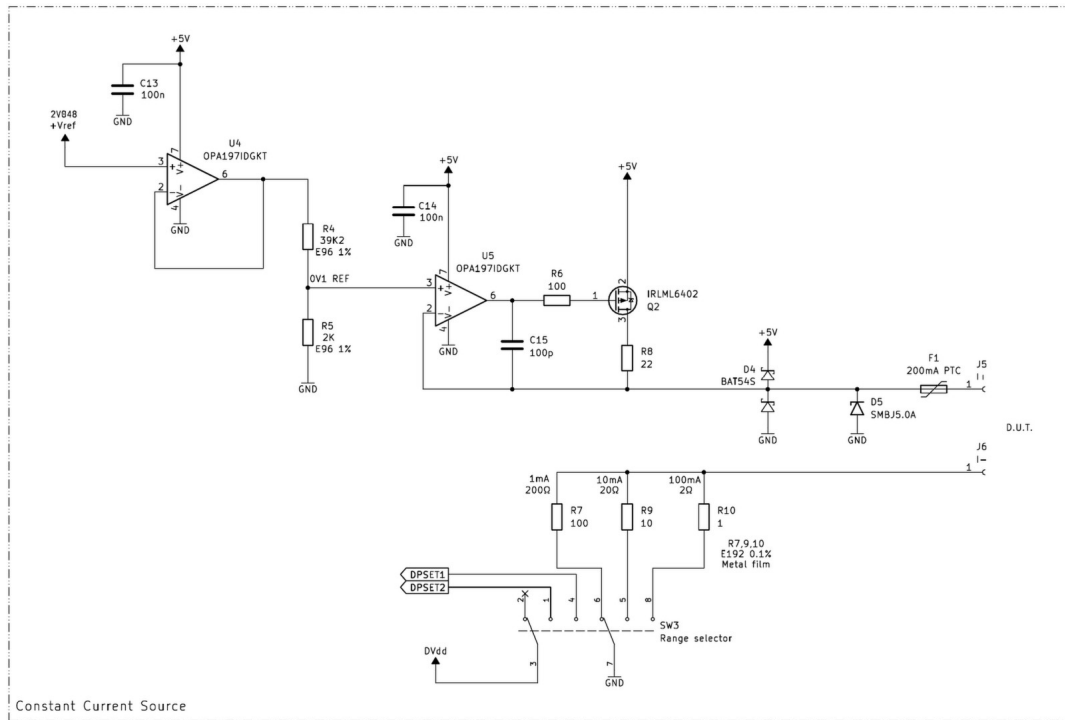


Figure 3: Constant Current Source

3.3 Input element

The input stage of the device consists of two S16N-PC jacks, input protection, an INA826 instrumentation amplifier, and an RC filter. The jacks (J7 and J8) serve as the voltage sense terminals of the device and are connected to the ADC inputs via the INA826 instrumentation amplifier (U7).

Input protection is implemented through a pair of PTC fuses (F2 and F3), a two-channel bidirectional TVS diode TP2E001DRLRG4 (U6) that protects the inputs from ESD, and two pairs of BAT54S diodes (D6 and D7) that provide overvoltage protection. The inputs of the instrumentation amplifier (U7) are protected against overcurrent by resistors (R11, R19 and R12, R20) in a manner that does not significantly affect measurement accuracy due to the low input bias current of the INA826 (U7). The instrumentation amplifier (U7) is configured to amplify the voltage signal with a range up to 200 mV to 2 V, for which the ADC is calibrated.

The RC filter formed by the combination of a resistor (R14) and a capacitor (C17) serves to eliminate external noise that could affect the measured signal. This is achieved by attenuating high frequencies typical of external interference.

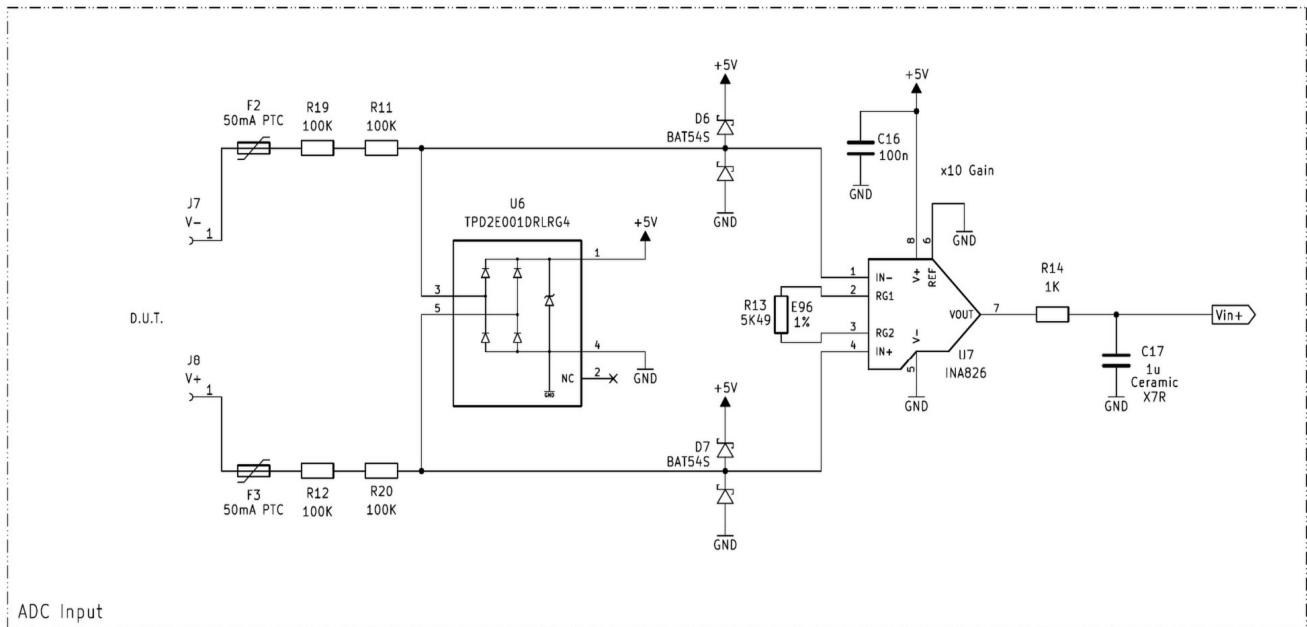


Figure 4: Input element

3.4 Continuity tester

The continuity tester is useful when diagnosing circuits. The indicator threshold is adjustable from 1 to 50 Ω and features both audio and visual indication.

A MAX9031 comparator (U9) is used to detect a "short circuit" using an adjustable reference voltage. This comparator features an integrated 4 mV hysteresis, which prevents output oscillation near the comparison threshold.

The reference voltage is implemented via a voltage divider consisting of a resistor (R16) and a potentiometer (RV1). The voltage divider is configured for a range corresponding to 1 to 50 Ω . By adjusting the potentiometer, the user can tune the comparator's voltage threshold.

Audio indication is provided by an active buzzer (BZ1) that emits a high-frequency tone upon detecting a "short circuit". Visual indication is provided by a red LED (D8).

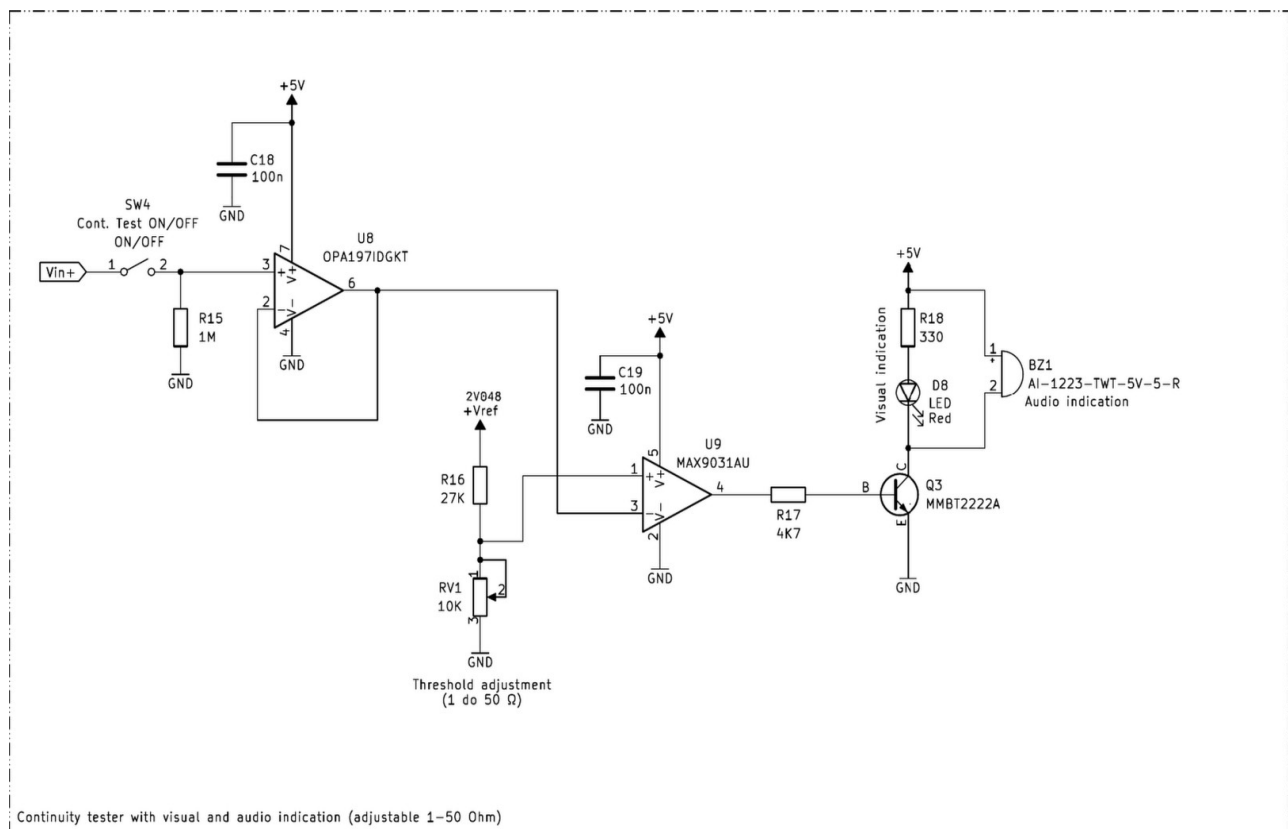


Figure 5: Continuity Tester

3.5 ADC circuit

A MAX1491 (**U1**) is used, which is an ADC with integrated LCD drivers offering 3½-digit resolution. This ADC provides high measurement precision and enables easy display of results on an LCD screen. Its low power consumption makes it suitable for battery-powered measuring devices. It is set to a 2 V range via the RANGE pin (pin 10) connected to ground, which corresponds to the output of the instrumentation amplifier.

A HOLD function is provided via a push button (**SW1**). This feature allows the user to freeze the displayed measurement result on the LCD screen at the press of a button. Low battery voltage indication is implemented using a voltage divider (**R1**, **R2**). When the battery voltage drops below 4.7 V, the LOWBATT segment lights up on the LCD screen.

The LCD outputs of the MAX1491 are multiplexed using triplex logic with a $\frac{1}{3}$ duty cycle signal. This approach significantly reduces the number of pins required to interface with the LCD screen. The LCD outputs are connected to the LCD connector (**J1**).

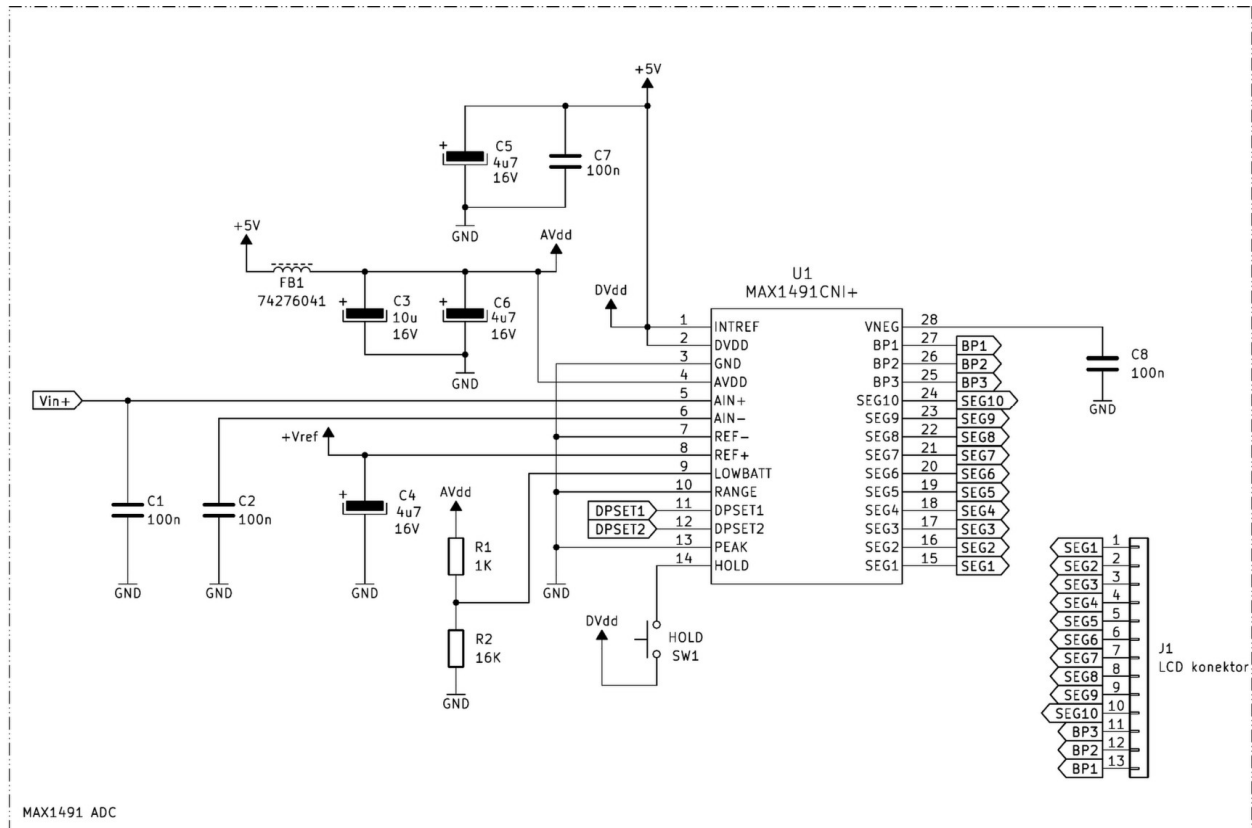


Figure 6: ADC circuit

3.6 LCD

The LCD screen (**U2**) is mounted on a separate PCB for greater flexibility when assembling it into the enclosure. The LCD connects to the MAX1491 via LCD connectors (**J1** and **J2**). The connector used is a Hirose DF13-13P. The LCD used is the VIM-503-DP-RC-S-HV. It is a static TN panel with multiplexed inputs, enabling straightforward interfacing with the MAX1491. The LCD panel itself features a 4½-digit display, of which 3½ digits are used to match the output of the MAX1491. Selecting this LCD represents a trade-off regarding project complexity; while a 3½-digit LCD with the same triplex multiplexing scheme is theoretically possible, it was not off-the-shelf available and would require custom manufacturing. Choosing this display avoids the complexity of adapting a passive display panel to the MAX1491 output logic.

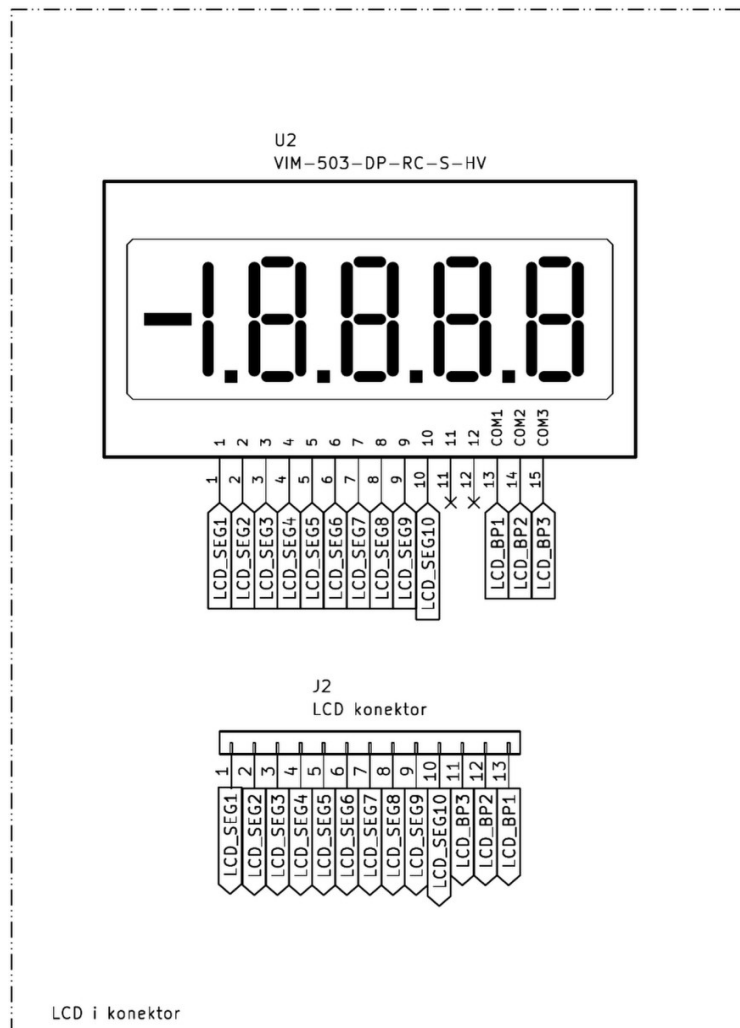


Figure 7: LCD & Connector

4 PCB

4.1 Main board

The board is constructed with two layers, with components placed on the top layer. Both layers feature a copper pour tied to ground (**GND**). This simplifies ground routing and minimizes the amount of copper that needs to be etched off during production. It also improves electromagnetic stability, making the device more resistant to noise.

Trace widths on the board were chosen to guarantee proper functionality and ensure physical durability:

- **50 mils (1.270 mm)** for power supply traces
- **20 mils (0.508 mm)** for signal traces

The aforementioned input protections are located away from other circuitry to prevent potential damage from high temperatures, overcurrent, or arcing. It is critical that these protective components be placed close to the input jacks for rapid fault response (especially PTC fuses **F1**, **F2**, and **F3**).

The voltage sense terminals are positioned between the current terminals with clear spacing to naturally discourage improper connections by the user. Color-coded terminals further reinforce this setup: the I+ terminal is red, while I- is black. The device is designed for use with specialized cables known as Kelvin clamps, which connect to I+ and V+, and I- and V-, respectively.

The device controls are strategically positioned along the edges of the board. The power switch is located on the rear panel, while the input jacks, HOLD button, and range selector are on the front panel. The short-circuit indicator switch and its threshold adjustment potentiometer are positioned on the right side.

All components on the board are grouped by circuit function and clearly labeled on the silkscreen. This significantly simplifies cross-referencing schematic functional blocks with physical board components. The PCB is designed primarily around SMD (Surface Mount Device) components. Compared to THT (Through Hole Technology), SMD components allow for a significantly smaller form factor, enabling the main board to fit within a 100x100 mm footprint. Compact physical dimensions are vital here, as a battery-powered device must remain portable and easy to handle.

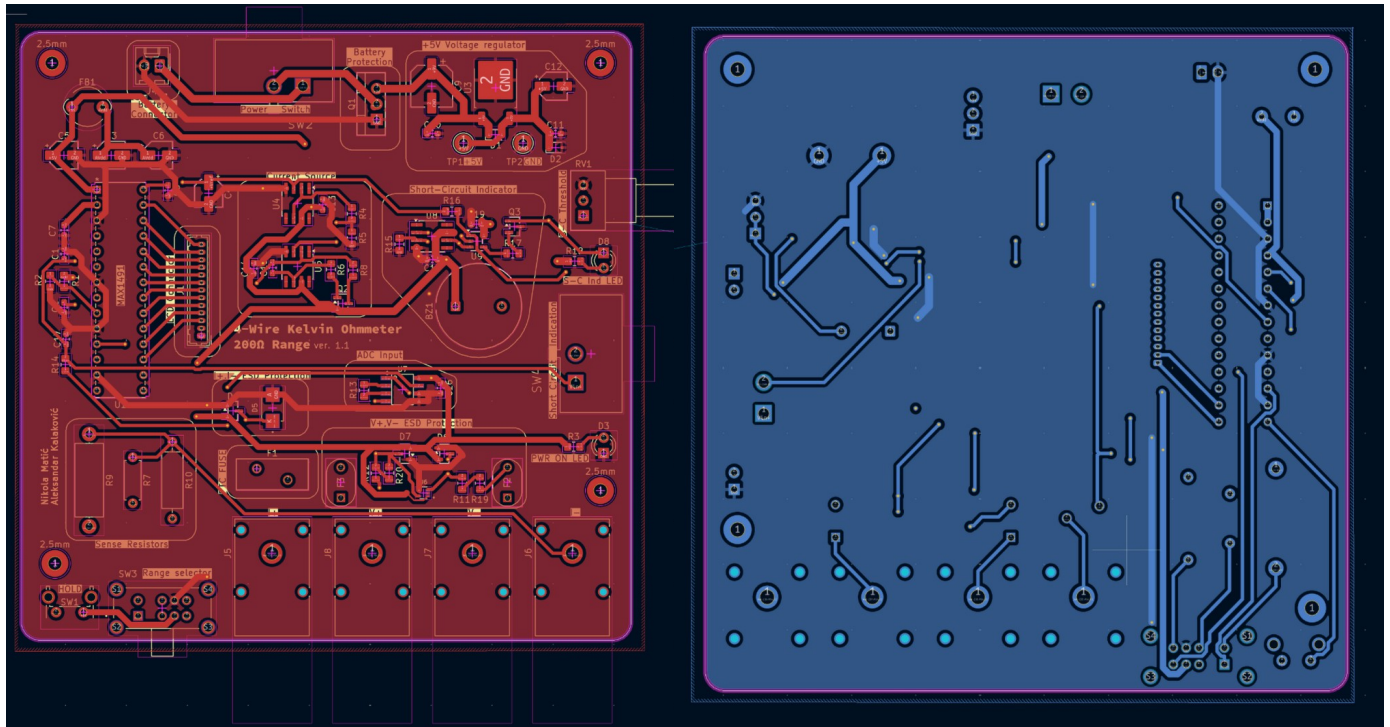


Figure 8: Front & back of the main board

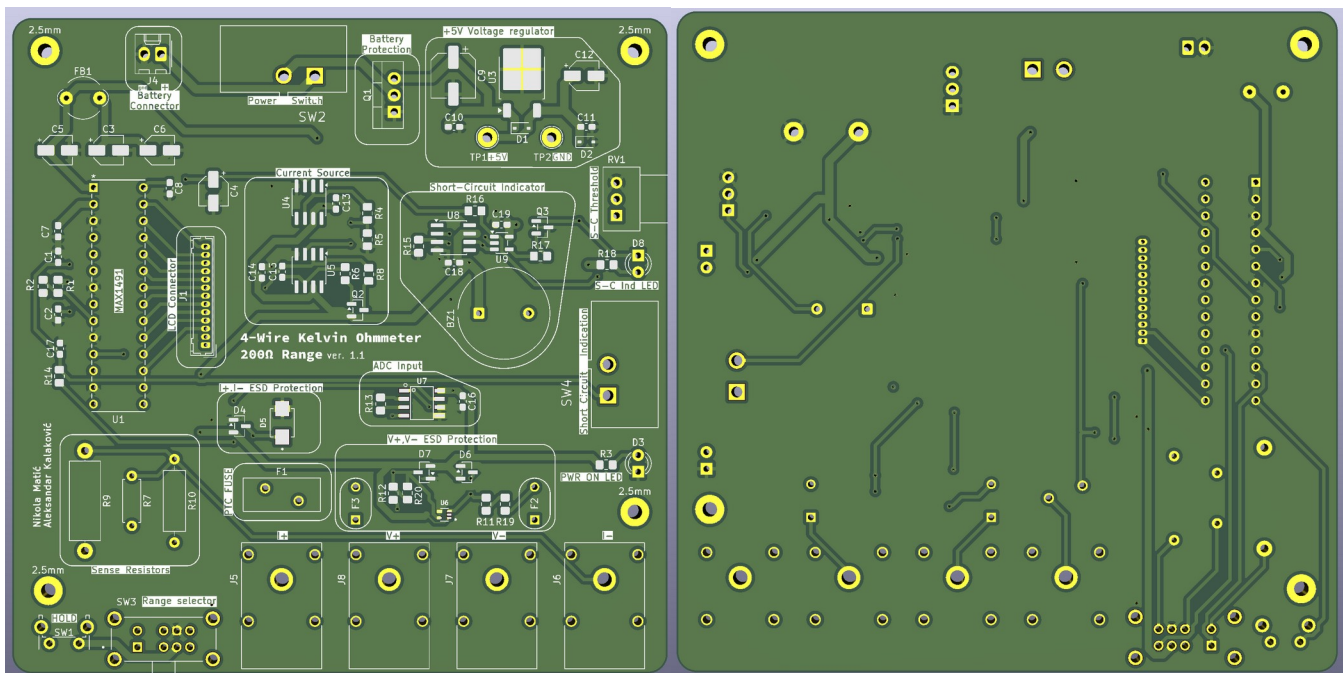


Figure 9: 3D projection of the main board (without components)

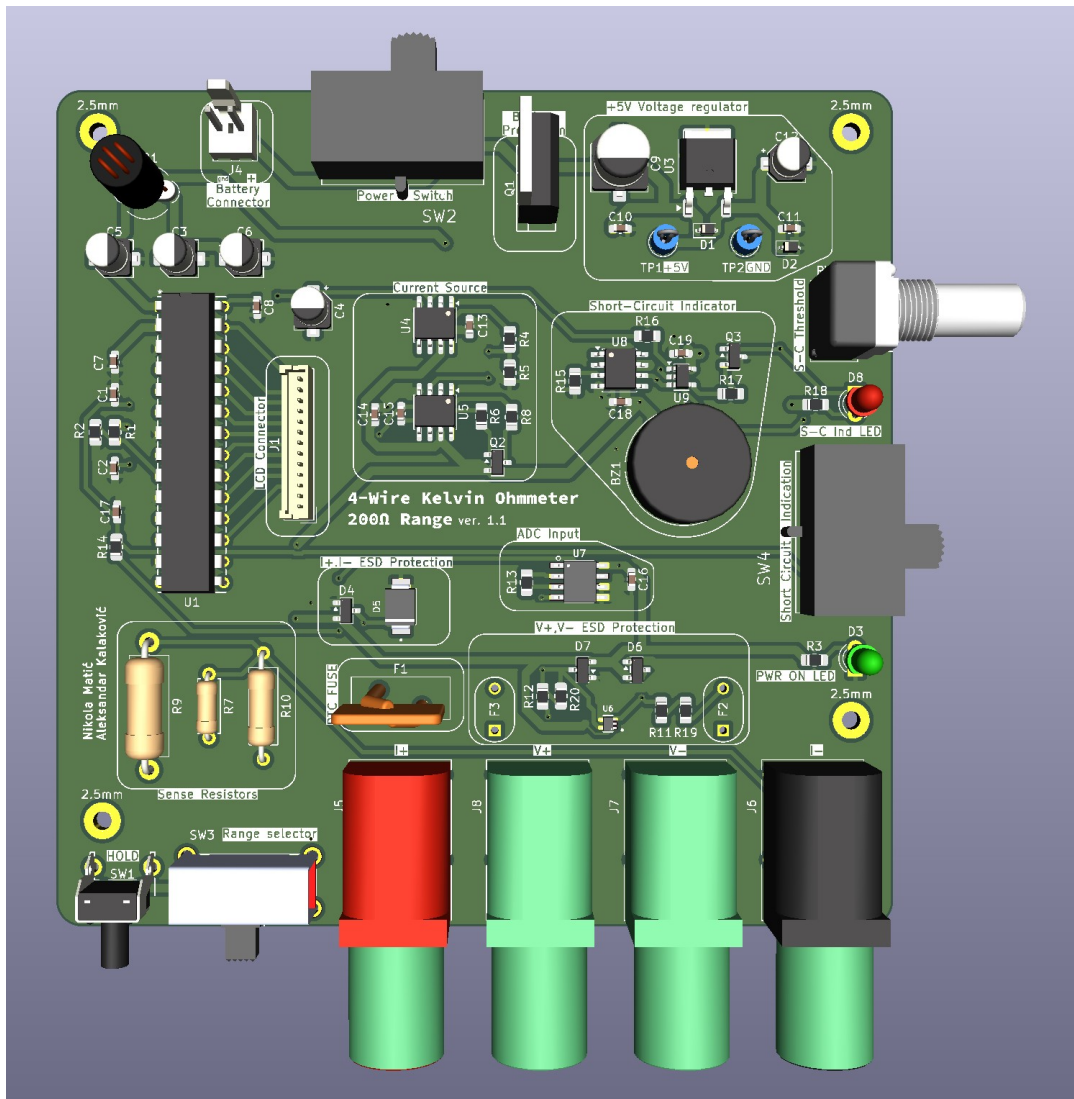


Figure 10: Main board with components

4.2 LCD board

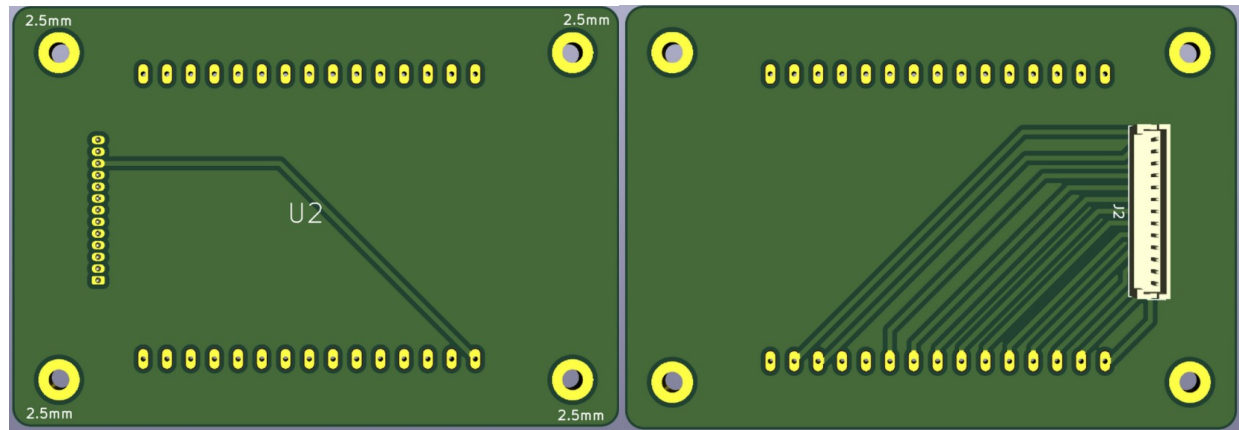


Figure 11: Front & back of the LCD board

4.3 Battery board

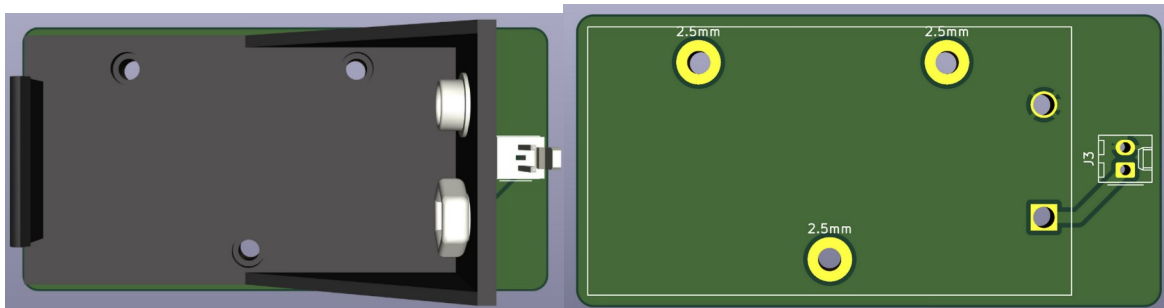


Figure 12: Front & back of the battery board

5 Enclosure



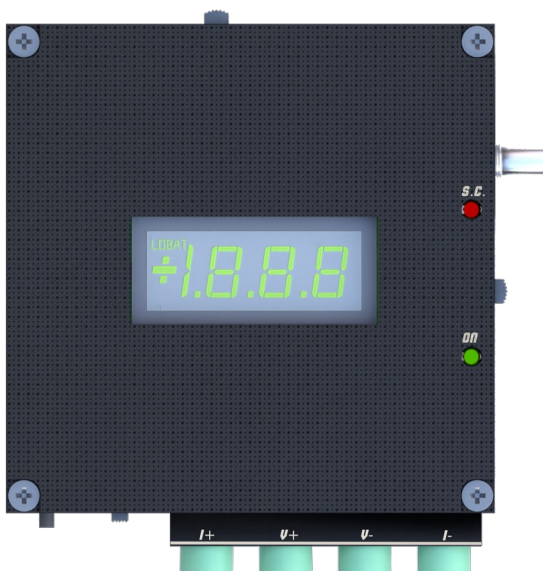
(a) Front



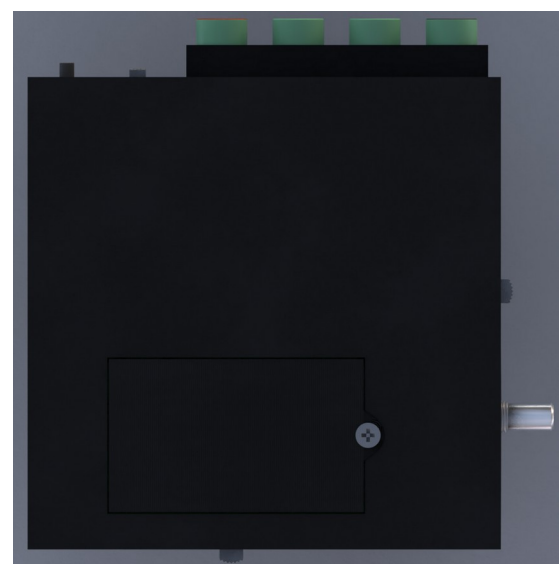
(b) Back



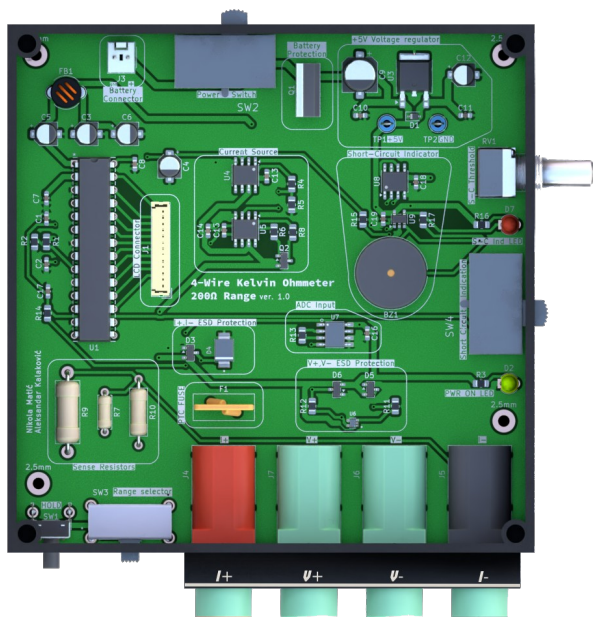
(c) Side



(d) Top



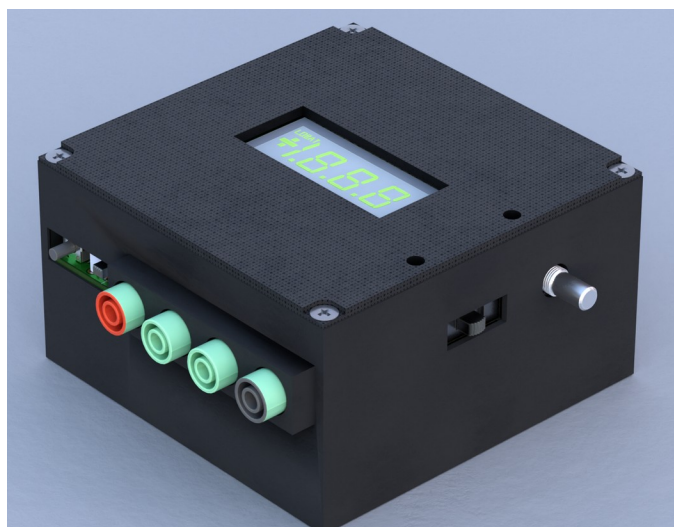
(e) Bottom



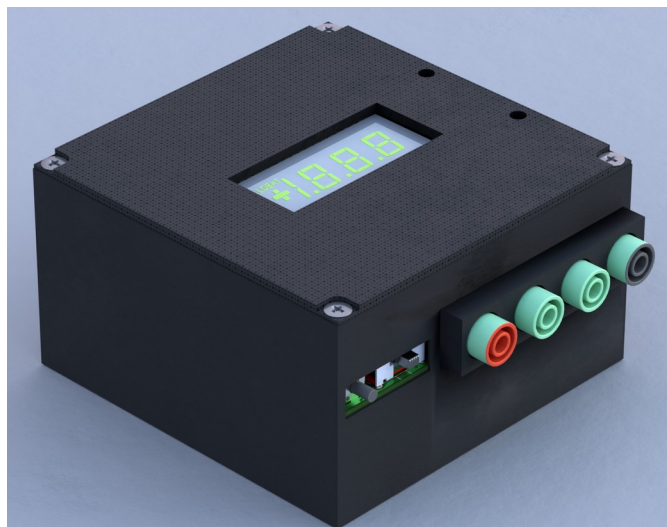
(f) Open top



(g) Open bottom

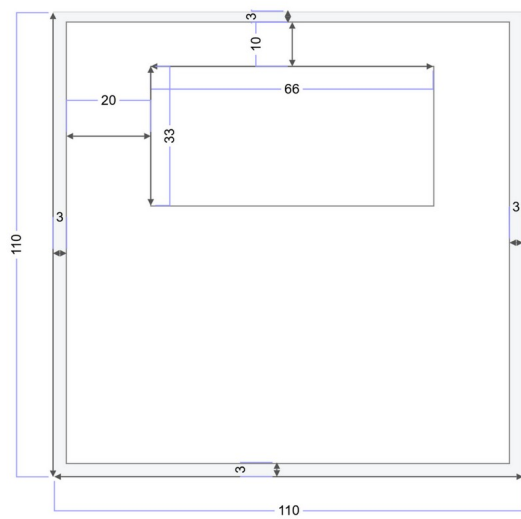


(h) 3D view 1

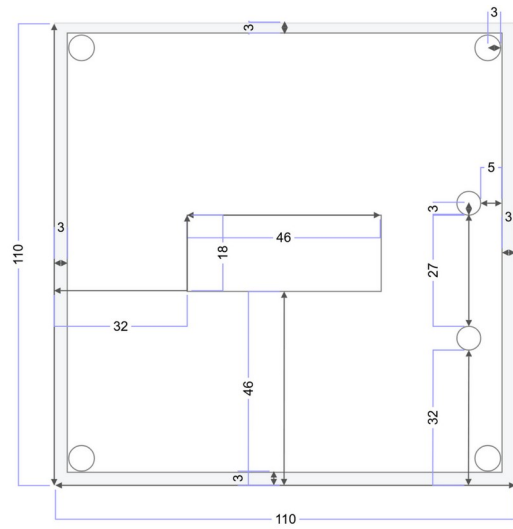


(i) 3D view 2

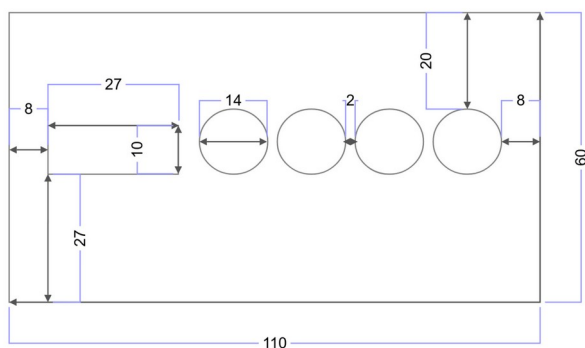
Figure 13: 3D projection of enclosure



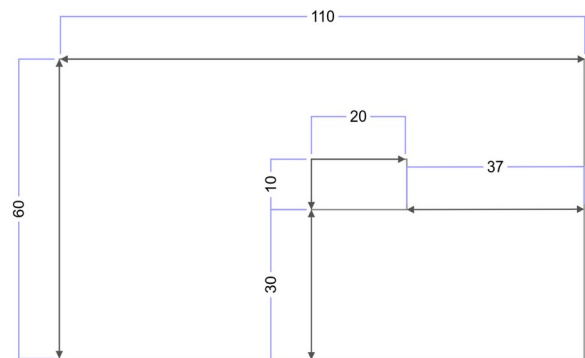
(a) Bottom



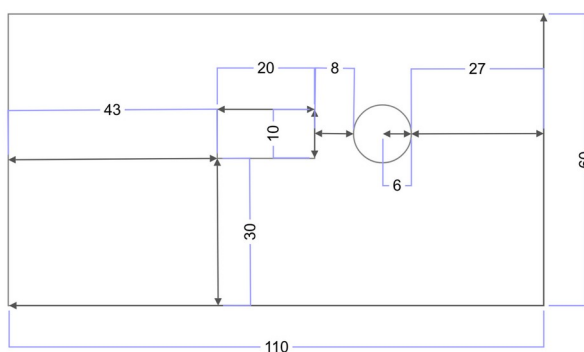
(b) Top



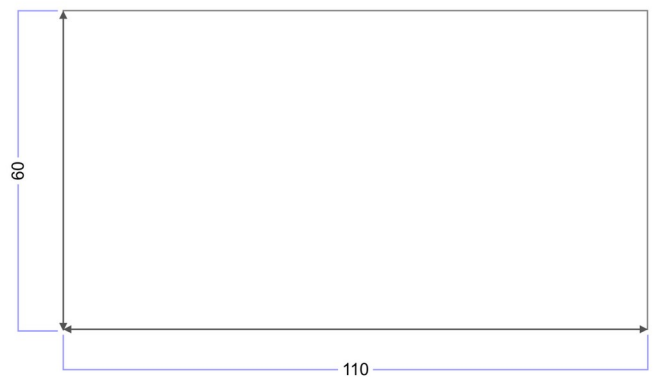
(c) Front



(d) Back



(e) Right side



(f) Left Side

Slika 14: Enclosure dimensions

6 BOM

Item	Value	Reference	Footprint	Qty	Mfr. part No.	Price (piece)(€)	Price (€)
1	Battery	BT1	12BH611-GR	1	12BH611-GR	1,46	1,46
2	Buzzer	BZ1	AI-1223-TWT-5V-5-R	1	AI-1223-TWT-5V-5-R	1,35	1,35
3	100n	C1, C2, C7, C8, C10, C11, C13, C14, C16, C18, C19	_0603	11	GCM188R71H104KA57D	0,03	0,33
4	10u	C3	4x5.4	1	EXV106M035A9DAA	0,43	0,43
5	4u7	C4, C5, C6	4x5.4	3	EXV475M035A9BAA	0,37	1,11
6	100u	C9	6.3x5.9	1	MAL214699805E3	1,79	1,79
7	1u	C12	4x4.5	1	EDH105M050A9BAA	0,16	0,16
8	1u	C17	_0603	1	0603X7R105KT7AT	0,49	0,49
9	100p	C15	_0603	1	KGQ15ACG2A101JM	0,05	0,05
10	1N4448	D1	SOD-323	1	1N4448WS-HE3_A-08	0,55	0,55
11	Power ON	D2	LED_D3.0mm	1	SLX-LX3054HD	0,04	0,04
12	BAT54S	D3, D5, D6	SOT-23	3	BAT54S	0,12	0,36
13	SMBJ5.0A	D4	DIOM4336X265N	1	SMBJ5.0A	0,53	0,53
14	LED	D7	LED_D3.0mm	1	SLX-LX3054HD	0,04	0,04
15	PTC Fuse	F1	MF-RX020/72-0	1	MF-RX020/72-0	0,34	0,34
16	Ferrite Bead	FB1	28C0236-0JW-10	1	74276041	0,12	0,12
17	LCD konektor	J1, J2	Hirose_DF13-13P-1.25DSA	2	DF13-13P-1.25DSA(55)	1,01	2,02
18	Konektor baterije	J3, J4	KK-254_AE-6410-02A	2	22272021	0,18	0,36
19	I+,I-,V+,V-	J5, J6, J7, J8	FCR7350R_S16N-PC	4	FCR7350R_S16N-PC	4,9	19,6
20	FQP27P06	Q1	TO-220-3	1	FQP27P06	2,51	2,51
21	IRLML6402	Q2	SOT-23	1	IRLML6402TRPBF	0,39	0,39
22	MMBT2222A	Q3	SOT-23	1	MMBT2222A	0,1	0,1
23	1K	R1, R14	_0805	2	RMCF0805JG1K00	0	0
24	16K	R2	_0805	1	RC0805JR-0716KL	0,09	0,09
25	330	R3, R17	_0805	2	RK73B2ATTE331J	0,1	0,2
26	39K2	R4	_0805	1	CRCW201039K2FKEF	0,3	0,3
27	2K	R5	_0805	1	RC0805FR-7W2KL	0,1	0,1

28	100	R7	DIN0207	1	PTF56100R00BYEB	5,15	5,15
29	100	R6	_0805	1	RK73B2ATTE101J	0,1	0,1
30	22	R8	_0805	1	SR0805JR-7W22RL	0,1	0,1
31	10	R9	DIN0414	1	UPF50B10RV	4,5	4,5
32	1	R10	DIN0309	1	RS1/21R000BB12	19,44	19,44
33	100K	R11, R12, R19,R20	_0805	4	SG73S2ATTD104G	0,2	0,8
34	5K49	R13	_0805	1	RT1206FRE075K49L	0,1	0,1
35	1M	R15	_0805	1	TC0550F1004T5G	0,01	0,01
36	27K	R16	_0805	1	AC0805JR-0727KL	0,09	0,09
37	4K7	R18	_0805	1	CRCW08054K70JNEAIF	0,29	0,29
38	10K	RV1	RK09Y11	1	RK09Y11L0001	1,84	1,84
39	HOLD	SW1	PTS645VM83-2 LFS	1	PTS645VM83-2 LFS	0,34	0,34
40	ON/OFF, Indikacija K.S.	SW2, SW4	L101011ML04Q	2	L101011ML04Q	2,55	5,1
41	Odabir opsega	SW3	MHS233RA04	1	MHS233RA04	2,62	2,62
42	MAX1491CNI+	U1	21-0043D_28_MXM	1	MAX1491CNI+	15,24	15,24
43	LM78M05_TO252	U3	TO-252-2	1	LM78M05CDTX/NOPB	1,39	1,39
44	OPA197IDGKT	U4, U5, U8	SOIC-8	3	OPA197IDGKT	1,78	5,34
45	TPD2E001DRLRG4	U6	SOTFL50P160X60-5N	1	TPD2E001DRLRG4	0,77	0,77
46	INA826	U7	SO-8	1	INA826AID	3,52	3,52
47	MAX9031AU	U9	SOT-23-5	1	MAX9031AUK+T	1,39	1,39
48	VIM-503-DP-RC-S-HV	U2	VIM-503-DP-RC-S-HV	1	VIM-503-DP-RC-S-HV	5,66	5,66
49	50mA PTC	F2, F3	CMF_RL50A-0_BRN	2	CMF-RL50A-0	0,61	1,22
Total:							109,8

Table 2: BOM (Mouser 20.5.2026.)